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RESEARCH ARTICLE

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Finding what you don't know: Testing SDM methods for poorly known species

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Abstract

Aim: A limitation of species distribution models (SDMs) is that species with low sample sizes are difficult to model. Yet, it is often important to know the habitat associations of poorly known species to guide conservation efforts. Techniques have been proposed for modelling species' distributions from a few records, but their performance relative to one another has not been compared. Because these models are built and evaluated with small data sets, sampling error could cause severely biased sampling in environmental space. As a result, SDMs are likely to underpredict geographic distributions given small sample sizes. We perform the first comparison of methods explicitly promoted or developed for predicting the geographic ranges of species with very low sample sizes.

Location: North Carolina, USA.

Taxon: South Mountains Grey-cheeked Salamander (*Plethodon meridianus*).

Methods: Using the sparse, existing georeferenced records of *P. meridianus*, we built SDMs using a range of methods that previous researchers have argued should work for low sample sizes. We then tested each SDM's ability to accurately predict independent survey data that were not georeferenced prior to our study. We compared SDMs using omission error and AUC.

Results: Roughly half of the models successfully predicted survey records in the range centre, and all models had high omission error rates in the range exterior. In the range of interior or exterior, the 'ensemble of small models' technique produced SDMs with high omission error rates. Spatial filtering had a negligible impact on model performance. Most, but not all, models outperformed predictions using distance from known populations. Using one of the best-performing methods, we developed an improved range map of *P. meridianus*.

Main Conclusions: Geographically peripheral populations were difficult to predict for all SDMs, though some methods were clearly inferior for our data set. We recommend that when sample sizes are low, researchers use Maxent with species-specific model settings.

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KEYWORDS
omission error, range maps, salamanders, sampling bias, spatial autocorrelation, species distribution modelling

1 | INTRODUCTION

Species distribution modelling seeks to predict and describe species' geographic ranges (Peterson et al., 2011). Presence data are combined with environmental data to produce heat maps of habitat quality. Because species distribution modelling is limited by sample size and sampling biases, it may be difficult to implement with poorly known species, such as those that are rare or difficult to detect across their range. Rarity may be a result of observer bias or underlying biological causes. Biological causes of rarity include biotic interactions, dispersal limitation, habitat loss and demographic constraints (Flather & Sieg, 2007; Rabinowitz, 1981). Species may also be difficult to detect not because they are rare *per se* but because of environmental constraints on activity patterns that make them observable during brief time windows (e.g., O'Donnell & Semlitsch, 2015). Because species distribution models (SDMs) generally perform poorly with fewer than 30 presences, we treat 30 presences as our upper cutoff for limited sampling (Wisz et al., 2008). Despite difficulties with building SDMs using few presences, several methods have been promoted, and a recent influential monograph suggested that modelling distributions with low sample sizes remain one of the most pressing issues in contemporary habitat modelling (Guisan et al., 2017). SDMs built using few presences may poorly represent a species' distribution due to overfitting or parameter misestimation caused by sampling biases (existing records are not representative of the full environmental conditions over which a species occurs). In the case of sampling bias, a resulting SDM can exhibit strong statistical support but underpredict the geographic range (Peterson et al., 2011).

One approach for building SDMs given low sample sizes is to use the software Maxent (Phillips et al., 2006). Sheth et al. (2014) found that Maxent can accurately model species' distributions using fewer than ten presences. Maxent models built using small sample sizes have been improved by using spatial filtering of presences (Pearson et al., 2007). Spatial filtering involves defining a distance, such that no presences used to build an SDM is within a specified distance of each other. This reduces the sample size, but assuming that environmental data are spatially autocorrelated, filtering reduces sampling bias in environmental space (Boria et al., 2014).

Another suggestion has been to build an array of SDMs in Maxent using different model settings and to compare their performance (Galante et al., 2018). Maxent models are frequently evaluated based on the area under the curve (AUC), a statistic that evaluates a model's ability to rank presences over random points within the geographic area over which the model is calibrated. For low sample sizes, models are evaluated via jackknifing (a model is built on all but one presence point and used to predict the withheld point) because the sample size is limiting (Shcheglovitova & Anderson, 2013). Alternatively, the Akaike information criterion corrected for small sample sizes (AICc) can be used (Galante et al., 2018).

Another method is the 'ensemble of small models' (ESM; Lomba et al., 2010). This highly parametric approach involves building multiple SDMs using two environmental variables at a time with multiple algorithms, weighting SDMs based on a measure of model performance and then using weights to create a single ensemble model. For example, if one were to model a species' distribution using six variables and four algorithms, this would result in 15 combinations of variables multiplied by four algorithms (prior to ensembling). Poorly performing SDMs are typically eliminated before ensembling.

Beyond performing better than a random prediction (AUC > 0.5), SDMs should outperform predictions from a simpler, competing explanation, such as distance from other populations (Hijmans, 2012). Nearest distance to known populations can be used to locate populations while not directly capturing anything about the environment. A distance raster, estimated from one set of populations, may be useful to predict other populations (not included in the first set) since environments are spatially autocorrelated or because metapopulation dynamics result in populations being near one another.

The South Mountains Gray-cheeked Salamander (*Plethodon meridianus*) is found in the South Mountains of North Carolina, USA, and nearby hills. The South Mountains are nested within the Appalachian Mountains. Highton and Peabody (2000) described *P. meridianus* based on geographic patterns of allozyme variation; it is similar in appearance to the allopatrically distributed species *P. amplus*, *P. metcalfi* and *P. montanus*. Since its description, no ecological data have been published for *P. meridianus*, save for physiological measurements of six individuals (Clay & Gifford, 2018). In addition to having few documented presences (described below), the geographic range of *P. meridianus* is very restricted (Figure 1). Salamanders in the genus *Plethodon* are completely terrestrial and lack lungs, and they depend on specific environmental conditions for cutaneous respiration. When ground conditions are cool and humid, salamanders emerge from burrows to forage and reproduce, otherwise remaining underground to avoid desiccation. This aspect of their behaviour results in low detection probability (O'Donnell & Semlitsch, 2015). We use *P. meridianus* as our test case for comparing SDM methods. *P. meridianus* has one of the smallest geographic distributions of any member of its genus (IUCN 2020) and is patchily distributed within its range (Radomski unpub.).

We ask five questions about building SDMs with small sample sizes. (1) How do SDM techniques recommended for working with small sample sizes compare to one another? (2) Does the performance of techniques differ when predicting withheld presences from the range interior as opposed to geographically peripheral withheld presences? (3) Can these techniques outperform a model agnostic to environmental data (a distance raster)? (4) Can data sets

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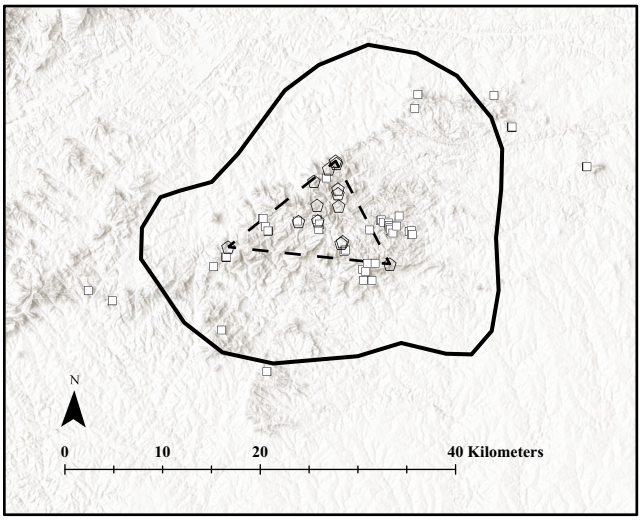

FIGURE 1 The IUCN range map of *Plethodon meridianus* is outlined in black. Hollow pentagons represent database presences, and white squares represent survey presences. The dotted triangle is the boundary for the interior versus exterior survey points

obtained from existing databases represent a biased region of environmental space within the greater environmental space that a species occupies? (5) How does changing the presence data set (database, surveys, or the combination of both) affect SDM predictions and complexity using the best-performing method?

2 | METHODS

2.1 | Database presences and environmental data

To obtain presences to build SDMs, we used records from VertNet, the Global Biodiversity Information Facility, and the Bell Museum of Natural History (Appendix S1). Working with environmental data at a coarse resolution (1 km ×1 km) would have resulted in many presences in the same grid cell, making many 'duplicate' presences that could not be used to build the SDM. To avoid this problem, we used environmental data at a high resolution (80 m × 80 m). Smaller grid sizes may increase the influence of spatial autocorrelation of environmental variables (Qi & Wu, 1996), though the topographic complexity of mountains justifies the use of higher resolutions, as montane regions exhibit lower spatial autocorrelation at high resolutions (Chevalier et al., 2021). The variables included in the model were elevation, slope, aspect, distance to the nearest stream, and per cent canopy cover (www.nconemap.gov.datasets; https://www.mrlc.gov/data?f%5B0%5D=category%3ATree%20Canopy). Slope and aspect were calculated from a digital elevation map in ArcGIS (ESRI, 2019). Aspect was transformed so that values ranged from 0° to 180° and were symmetrical across the *y*-axis. Canopy cover can change drastically over time from clear-cutting or wildfires. Forests of the South Mountains have not been recently clear cut, though there was a large wildfire in 2016, which is the same year that canopy cover was measured. Upon inspecting georeferenced points from this study, we found that they do not correspond to burned sites, as burned sites were mostly on dry ridges. Canopy cover data were upscaled from 30 m × 30 m to 80 m × 80 m. All five environmental variables are important predictors of ground temperature and humidity in the southern Appalachian Mountains (Fridley, 2009) and therefore important for terrestrial salamander distributions (Gade & Peterman, 2019). No variables were highly correlated ($r < 0.6$), so we used all five of them. Rasters are provided in Appendix S2.

We built the calibration area by buffering the IUCN range map (Figure 1, IUCN 2020) by 2 km. Using a larger calibration area would risk sampling the range of *P. amplus*, *P. metcalfi*, or *P. montanus*, which may exclude *P. meridianus* through competition, dispersal limitation, or hybridization (Heidler, 2020). Furthermore, those areas are far enough away that *P. meridianus* may not have been able to access those habitats due to dispersal limitations. We eliminated duplicate presences in the same raster cell. We removed three presences that fell outside of the currently recognized range map for *P. meridianus*, though we note that IUCN range maps can have errors (Brooks et al., 2019). To be conservative, we treated the three presences as erroneously georeferenced and excluded them when building SDMs. One point corresponded to an agricultural field, which is unlikely habitat given the natural history of *Plethodon*. The other two points occurred on the other side of the Catawba River; rivers form important boundaries for *Plethodon* distributions in Appalachia (Highton, 1995). As none of the documented range was north of the Catawba River, we conservatively assumed those two points were erroneously georeferenced. After removing duplicates and presences outside the IUCN range map, we had 14 total presences.

2.2 | Building SDMs

We built 12 SDMs using different techniques (Table 1, Question 1). For our first set of SDMs (a–d), we used all 14 presences. These were analyzed (a) in Maxent using default settings, (b) in Maxent with parameter values assessed through model comparison using AICc, (c) in Maxent with parameter values assessed through model comparison using AUC and jackknifing presences and (d) the ESM. To build models b and c, we compared models by changing the regularization multiplier from 1.0 to 5.0 in increments of 0.25. The regularization multiplier penalizes model complexity, such that larger values encourage the use of fewer parameters. We also varied the number of feature classes used by Maxent with the following options: linear, linear + quadratic, hinge, linear + quadratic + hinge, linear + quadratic + hinge + product and linear + quadratic + hinge + product + threshold. The ESM was built by making bivariate SDMs using artificial neural networks and boosted regression trees; these two algorithms performed well when used in ESMs (Breiner et al., 2018) and attempts to include several other algorithms failed. Bivariate SDMs were evaluated via jackknifing and calculating AUC. We ensembled bivariate SDMs if their AUC score exceeded 0.70 and then weighted by

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TABLE 1 Predictive ability and spatial autocorrelation of SDMs

Model	Method	Spatially Filtered	Distance Raster	$OR_{survey1}$	$OR_{survey2}$	$OR_{survey3}$	$AUC_{survey1}$	$AUC_{survey2}$	$AUC_{survey3}$	Moran's I_1	Moran's I_2
a	Maxent – default	No	No	0	0.542	0.406	0.962	0.805	0.793	0.069	0.228*
b	Maxent – tuning – AICc	No	No	0	0.542	0.406	0.961	0.810	0.812	0.053	0.224*
c	Maxent – tuning – AUC	No	No	0	0.625	0.469	0.961	0.789	0.797	0.012	0.204*
d	ESM – AUC	No	No	0.375	0.912	0.78	0.955	0.811	0.812	0.074	0.109
e	Maxent – default	Yes	No	0	0.583	0.438	0.961	0.784	0.808	0.035	0.228*
f	Maxent – tuning – AICc	Yes	No	0	0.625	0.469	0.961	0.788	0.796	0.032	0.223*
g	Maxent – tuning – AUC	Yes	No	0	0.625	0.469	0.961	0.788	0.796	0.032	0.204*
h	ESM – AUC	Yes	No	0	0.583	0.438	0.954	0.595	0.662	0.088	0.213*
i	ESM – AUC	No	Yes	1	1	1	0.941	0.811	0.809	0.039	−0.026
j	ESM – AUC	Yes	Yes	1	1	1	0.939	0.589	0.649	0.057	0.210*
k	ESM – BI	No	No	1	1	1	0.664	0.607	0.606	0.094	−0.027
l	ESM – BI	Yes	No	0.125	0.625	0.531	0.855	0.663	0.687	0.009	−0.027
DR	NA	NA	NA	1	1	1	0.950	0.691	0.720	NA	NA

Note: Methods labelled as 'Maxent - default' use the default regularization parameter (1.00) and feature classes (linear + quadratic + hinge + product + threshold). 'Maxent - tuning' indicates that the regularization parameter and feature classes were empirically chosen using model selection. Under 'Method', the statistic used for model selection is indicated where applicable (AICc, Akaike's information criterion corrected for small sample zize; AUC, area under the curve; BI, Boyce index). In the 'Model' column, 'DR' indicates the distance raster. $OR_{survey1}$ and $AUC_{survey1}$ represent models' ability to predict survey presences in the range interior; ORs_{urvey2} and $AUC_{survey2}$ represents models' ability to predict survey presences in the range exterior; and $OR_{survey3}$ and $AUC_{survey3}$ represents models' ability to predict survey presences in the entire area. Spatial autocorrelation was measured by Moran's I_1 in the range interior and by Moran's I_2 in the range exterior. Asterisks indicate significant spatial autocorrelation.

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AUC before ensembling. A threshold AUC of 0.70 has conventionally been treated as the minimum value of an adequate model (Peterson et al., 2011). The individual algorithms in the ESM were built using model tuning, which changes model settings to improve the desired measure of statistical performance (AUC in this case). We repeated the previously described procedures using a spatially filtered data set of presences (e–h). Spatial filtering was done using the R package 'spThin' (Aiello-Lammens et al., 2015), where presences had to be at least 0.5 km apart, resulting in 11 presences. The filtering distance of 0.5 km accomplished our goal of reducing spatial bias while retaining most database presences. We built two more models with the ESM approach using the (i) unfiltered and (j) filtered presence data with a spatial distance layer (nearest neighbour distance) as an additional predictor. The spatial distance layer represents Euclidean distances to the nearest presence (calculated from the 14 presences). Allouche et al. (2008) found that thresholded distance rasters could be combined with thresholded SDMs to improve the prediction of undocumented populations, though this was less true for species with few presences. We directly incorporated a distance raster into the model-building process. We were curious whether it would improve SDM predictions when sampling was sparse because it may encourage models to recognize that locations near known presences are more likely to contain unknown populations by acknowledging that organisms' distributions are spatially autocorrelated. We attempted to incorporate the distance layer into Maxent models, but this resulted in SDMs that only used information from the distance layer and thus predicted suitable habitat at the exact location of presences used to build the model and predicted no other suitable raster cells. We also attempted to incorporate a binary thresholded distance raster into Maxent models, where the threshold was equal to the maximum nearest neighbour distance among presences. This resulted in Maxent predicting zero suitability everywhere beyond the threshold, which we regard as uninformative. Due to these problems, we do not consider Maxent models incorporating distance in any detail. Finally, an additional two models (k and l) were built using the ESM method, but instead of retaining models by AUC score, we used a threshold value of the Boyce Index (Boyce et al., 2002; Hirzel et al., 2006). While AUC measures model performance on the ability to discriminate presences over pseudoabsences, the Boyce index emphasizes the ability of models to predict a greater frequency of presences in areas of greater habitat suitability. Models k and l were built using the full and spatially filtered data sets, respectively, without the distance raster (inclusion of the distance raster when calculating ESMs caused errors when using the models thresholded by the Boyce index). We chose a threshold of 0.5. While the Boyce index has not been as widely adopted as AUC, some researchers have argued for its superiority as a metric for evaluating SDMs (Guisan et al., 2017). However, there is no 'perfect' measure of SDM performance. Maxent models were built using the R package 'ENMeval' (Muscarella et al., 2014), and ESMs were built using the R package 'ecospat' (Cola et al., 2017). All models were built using 10,000 pseudoabsence points (Barbet-Massin et al., 2012). We used pseudoabsences (instead of true absences) to build SDMs because collecting absence data with terrestrial salamanders is difficult due to their low detection probability (O'Donnell & Semlitsch, 2015).

AUC has been challenged as a useful metric of model performance (e.g., Lobo et al., 2008; Peterson et al., 2011). First, AUC equally weighs omission and commission errors (false negatives and false positives, respectively). Given that there are typically many more background points than presence points when modelling species' distributions, this can lead to an inflated measure of model performance. Omission errors are generally considered to be more problematic than commission errors (Anderson et al., 2003). Second, while discrimination ability is a convenient measure of model performance, it gives no indication of how well a model fits the data. Third, the meaning of discriminating presence points over 'background' or 'pseudoabsence' points has an unclear meaning, since the latter can result from a true absence or a lack of detection. Fourth, AUC is sensitive to sample size, such that species with few presence points relative to the area over which a model is calibrated will have an inflated AUC score. AUC may still be useful for selecting optimal model parameterization among competing models when models are calibrated and tested within the same time and space (Galante et al., 2018).

2.3 | Survey data

Our survey data were collected independently as part of three projects by different authors of this study (Appendix S1). In the first project, searches were conducted across the South Mountains during Spring 2011–2017. Efforts varied by site and year, but the primary methodology involved searching for *P. meridianus* during evening hours or, less commonly, by overturning stones and rotten logs during daytime hours. A total of 38 sites were sampled, with positive detection of *P. meridianus* at 19 sites. In the second project, elevational transects were conducted to search for *P. meridianus* in the South Mountains in Fall 2016. Searches for *P. meridianus* were conducted over steep transects with the goal of finding the lower elevational limit of *P. meridianus*, spanning the approximate range of 300–900 m in altitude. In the third project, searches were conducted within South Mountains State Park in May 2019 and September 2019. Survey data were haphazardly collected before SDMs were built; we did not conduct stratified sampling informed by SDM predictions (e.g., Rosner-Katz et al., 2020). However, interpreting results from stratified sampling for organisms with low detection probability is difficult because they risk a high false-negative rate.

Of all survey presences, none were at the same locations as those represented by database records (Figure 1). We eliminated duplicate survey presences in the same raster cell. Because survey presences were spatially clustered (Figure 1), we filtered them at a minimum distance of 0.5 km (the same filtering distance used for database presences) prior to using them to test SDMs. For the purpose of testing SDMs, we placed survey presences into three groups: 'survey1'

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were points falling within a minimum convex polygon formed by database presences, 'survey2' were points falling outside of the same minimum convex polygon and 'survey3' were the combination of both (Figure 1).

We tested whether the database records were in a different region of environmental space than the survey records (Question 4). Output from Maxent suggested that elevation was the most important explanatory variable of the distribution of *P. meridianus* (see 'Results'). We used the Mann-Whitney *U* test to compare the elevations of the presences used to build models and the survey data.

We evaluated the multivariate environmental similarity surface (MESS; Elith et al., 2010). The MESS is a map that describes where models are extrapolating. Negative values of MESS indicate extrapolation outside the range of predictors during model calibration, which could have negative consequences on model performance (Elith et al., 2010). All SDMs were projected onto an area including the five counties that contained all known populations of *P. meridianus*: Burke, Catawba, Cleveland, McDowell and Rutherford. We visually inspected the MESS to evaluate whether extrapolation may have impacted SDM predictions.

2.4 | Model comparisons

We evaluated SDM performance by comparing them against each other (Question 1). We compared models by calculating the 10% omission rate (OR), which is calculated by thresholding an SDM by the value that would exclude ten per cent of training presences. If more than 10% of survey presences are in an 'unsuitable' habitat, then the SDM has poor predictive power, likely due to overfitting (Galante et al., 2018). Survey presences do not include any of the presences obtained from databases. Some survey presences were within the area circumscribed by database presences, while other survey presences were outside this area (Figure 1); we tested the ability of SDMs to predict surveyed presences separately ($OR_{survey1}$ versus $OR_{survey2}$) and together ($OR_{survey3}$) because spatial autocorrelation will vary with distance and topography. First, $OR_{survey1}$ was calculated based on each model's ability to predict survey points in the range interior of *P. meridianus* ($N = 8$). The range interior is defined as the area bounded by a minimum convex polygon from database presences (Figure 1). Second, we calculated $OR_{survey2}$ by testing each model's ability to predict survey presences outside of the minimum convex polygon ('range exterior'; $N = 24$). Finally, we calculated $OR_{survey3}$ for all survey points ($N = 32$). As an additional evaluation of SDMs, we calculated AUC_{survey} for the three sets of survey presences. To calculate AUC_{survey}, 10,000 random points were generated within the area used to test models (Barbet-Massin et al., 2012). For $AUC_{survey2}$, we generated random points from an area consisting of the IUCN range map plus a corresponding buffer area of 6 km around all presences from this study. Prior to generating random points, we visually inspected different buffer areas; we found that this area was large enough to capture areas that are likely poor habitat yet also exclude suitable habitat to which *P. meridianus* likely cannot disperse (Barve et al., 2011). For $AUC_{survey3}$ (using all survey points), background points were sampled from the entire area (within and outside of the minimum convex polygon). We tested for a correlation between $AUC_{survey1}$ and $AUC_{survey2}$, and $OR_{survey1}$ and $OR_{survey2}$. A positive correlation between either pair indicates that SDM methods that perform better in the range interior also perform better in the range exterior; a negative correlation indicates a tradeoff in performance (Question 2). We also compared the predictive ability of the distance raster in isolation to each SDM ('DR' in Table 1, Question 3).

We interpreted values of AUC_{survey} as 'very good' if $AUC_{survey} > 0.9$, 'good' if $0.9 > AUC_{survey} > 0.8$ and 'moderately good' if $0.8 > AUC_{survey} > 0.7$ (Peterson et al., 2011). While we could additionally use the Boyce index to evaluate models, this may not be appropriate in our application of modelling. This is because most of the survey data were collected to identify geographic range margins of *P. meridianus*. Range margins frequently correspond to a biased region of low habitat suitability (Lee-Yaw et al., 2016; see 'Results').

While AUC is often considered a dubious indicator of SDM performance, we see it as complementary to OR because AUC is a measure of relative model performance (comparing SDM predictions for presences versus a random sample of the environment), while OR requires one to turn continuous models into binary predictions. An SDM could have a high AUC value, indicating strong discrimination ability, even though it is simply discriminating presences over background points in a geographic region where all habitat suitability predictions are low. This is useful because it indicates the SDM is predicting habitat suitability in the correct direction along a gradient of habitat suitability, yet an evaluation with OR would probably not predict any presences (OR = 1) because all sites in that geographic region have low suitability.

Using the same method used to build Model b, we compared model predictions using different sets of presence data (Question 5). We chose the methods for Model b (Maxent using model selection via AICc without spatial filtering) because it was one of the top-performing techniques for each test statistic (see 'Results', Table 1). We built an SDM using the sum of filtered survey and database presences (Model b_{all}; $N = 56$) and then the filtered survey presences (Model b_{survey}; $N = 32$). We compared these three SDMs by visual inspection and calculating niche overlap using Schoener's *D* (Schoener, 1968; Warren et al., 2008). Schoener's *D* range from 0 to 1, where 0 indicates no similarity and 1 indicates identical predictions. We calculated niche overlap after projecting each SDM to the extent of including all five counties. We used Maxent output to understand how variable importance changed with the presence data set used.

Because populations tend to be clustered in geographic space (a form of spatial autocorrelation) and environmental similarity is correlated with distance, there is valid skepticism that SDMs are ecologically meaningful (Fourcade et al., 2018). Journé et al. (2019) showed that when species' distributions are simulated without regard to any environmental variables, SDMs may still predict their ranges well. In the worst circumstances, significant positive spatial autocorrelation of SDM predictions might indicate that SDMs are simply capturing

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spatial autocorrelation of environmental variables that is coincident with, but not predictive of, the species' distribution. We tested for spatial autocorrelation in all of our SDMs using Moran's *I* (Moran, 1950). This was conducted in two regions of geographic space. The first (Moran's I_1) was calculated for habitat suitability at presences (including those used to build the model and survey presences) within the minimum convex polygon; these were spatially filtered by 0.5 km. The second (Moran's I_2) was calculated for habitat suitability at presences outside of the minimum convex polygon, also filtered at 0.5 km. Values of Moran's *I* were calculated using Euclidean distances among populations.

R code used for analyses is provided in Appendix S3.

2.5 | Improved range map

Given that some of our survey populations are outside the recognized geographic range of *P. meridianus*, we developed a new geographic range map using all presences (database and survey) with the method of Model b_{all}. We set a threshold equal to the minimum training presence, which ensured that all presences were part of the geographic range. We then calculated the intersection of this area with a 6 km buffer area generated from all presences of *P. meridianus*. We used a buffer area, as opposed to a minimum convex polygon because it is unlikely that our most peripheral presence data actually represent the exact range edge (Gaston, 1994). The buffer extends slightly beyond where *P. meridianus* were found during repeated field surveys, but given the potentially dynamic nature of range limits (Sexton et al., 2009), we believe this is appropriate. This resulted in two large polygons containing all known presences after sliver polygons were eliminated. We used the 'Smooth Polygon' tool in ArcGIS with Polynomial Approximation with Exponential Kernel using a smoothing tolerance of 100 km. We eliminated holes within the geographic range.

3 | RESULTS

3.1 | Model performance

In the range interior, all Maxent models, the ESM with AUC weighted models without spatial filtering, the ESM with AUC weighted models with spatial filtering and the ESM with Boyce Index weighted models with spatial filtering had low omission errors ($OR_{survey1} < 0.5$; Table 1; Question 1). In the range exterior, all models had high omission rates ($OR_{survey2} > 0.5$). When evaluating the entire range, all Maxent models and the ESM with AUC weighted models with spatial filtering had low omission errors ($OR_{survey3} < 0.5$). Nearly all models performed very well at discriminating presences over background points in the range interior ($AUC_{survey1} > 0.9$), with the exceptions being the ESM with Boyce Index weighted models without spatial filtering ($AUC_{survey1} = 0.855$) and the ESM built using the same criterion with spatial filtering ($AUC_{survey1} = 0.664$). All but four models

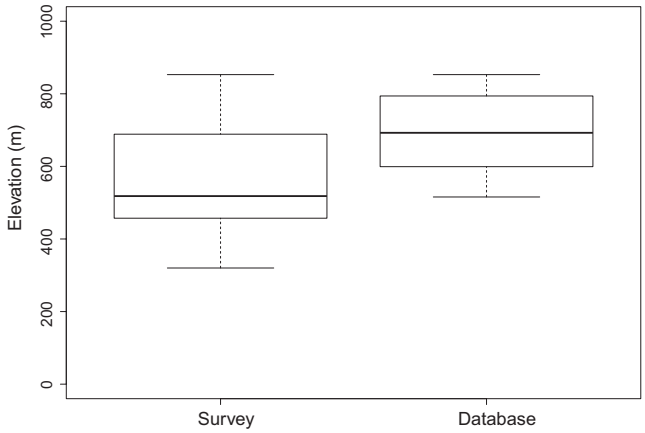

FIGURE 2 Box and whisker plot comparing elevation of surveyed populations and populations from publicly available databases. The difference was statistically significant, and the range of elevations in the survey data is much greater

performed moderately well at predicting surveyed populations outside the minimum convex polygon ($AUC_{survey2} > 0.7$). The four that performed poorly were variants of the ESM. All but four models, which were variants of ESMs, performed well at predicting the total of surveyed populations (the combined set of the range interior and exterior survey sites) (Table 1). The distance raster performed very well in the range interior ($AUC_{survey1} = 0.950$), poorly in the range exterior ($AUC_{survey2} = 0.691$) and moderately well for the combined data set ($AUC_{survey3} = 0.720$). Spatial filtering decreased omission errors for some ESMs, but the opposite was observed for Maxent. Adding information about distance appeared to increase omission errors for ESMs but had a negligible impact on AUC. Among Maxent models, spatial filtering improved model performance, though changes in AUC were modest. $AUC_{survey1}$ and $AUC_{survey2}$ were not correlated ($p = .06$, $\rho = 0.53$). $OR_{survey1}$ and $OR_{survey2}$ were correlated, but this is driven by many models having omission rates of zero or one. Thus, SDM methods do not seem to trade off accuracy in the range interior versus exterior (Question 2). Using AUC, the distance raster outperformed all ESMs in the range interior except models d and h; the distance raster outperformed all ESMs in the range exterior except models d and i (Question 3). However, the distance raster had high omission rates, which is a logical consequence of using geographic distance from populations to define habitat suitability under a threshold.

In the range interior, model predictions were not spatially autocorrelated using Moran's *I*, though outside the minimum convex polygon, all but four models (variants of ESMs) exhibited significant spatial autocorrelation (Table 1). In three cases (using ESMs) there was nonsignificant negative spatial autocorrelation of SDM predictions in the range exterior. Survey presences were, on average, at lower elevations than presences used to build the SDMs ($p = .016$), but survey presences encompassed the entire elevational range of database presences (Figure 2; Question 4). The MESS indicated that there was extrapolation when models were transferred to the area containing all five counties. However, this was driven by higher

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elevations northwest of where *P. meridianus* is found. This area is where the allopatric and morphologically similar species *P. montanus* is found. SDMs predicted high suitability in this area, due to the overwhelming influence of elevation on the SDM predictions.

Model b, which was regularly one of the top-performing techniques (Table 1), used Maxent with AICc as the criterion for optimal model selection without filtering presence data (Table 1; Question 5). With only database presences, settings chosen based on AICc used the linear feature class with a regularization multiplier of 1. Estimates of variable importance inferred that elevation explained 98% of model predictions; nearest distance to stream and aspect had no contribution. For Model b_{all} (using all spatially filtered presences), the optimal model settings used the hinge feature class with a regularization multiplier of 1.5. Elevation had the greatest impact (70%), but the importance of all other variables increased compared to when only using database presences, with canopy cover being 19% and nearest distance to the stream being 8%. Slope and aspect each contributed 1.5%. For Model b_{survey} (using spatially filtered survey presences), optimal settings were linear + quadratic feature classes and a regularization multiplier of 2. For Model b_{survey}, elevation explained 52% of predictions, canopy cover explained 39%, distance to nearest stream explained 9%, slope contributed <1% and aspect made no contribution. Model b made 'smooth' predictions, which is likely because it used only a linear feature class, and model predictions were almost entirely influenced by elevation (Figure 3a). Model b_{all} made much less smooth predictions (Figure 3b), probably because more variables were important, which could be identified using a larger sample size (56 observations versus 14). We suspect that since canopy cover had a large impact and it varies by land use, Model b_{survey} is patchy in appearance compared to Models b and b_{all} (Figure 3c). Model b made the most geographically narrow predictions of suitable habitat, appearing to be nested within the predictions of Model b_{all} and Model b_{survey} (Figure 3). After projecting the three SDMs onto the greater region consisting of five counties, we calculated niche overlap. Model b_{all}, which used all points, was intermediate to the other two model predictions. Niche overlap between Model b and Model b_{all} was 0.75, the overlap between Model b and Model b_{survey} was 0.71, and overlap between b_{all} and b_{survey} was 0.82.

The new range map we developed for *P. meridianus* was designed to capture all known presences and areas that could likely be inhabited by *P. meridianus*, but that may have gone undetected (Figure 4). Our new range map differs from the previous delineation of its geographic distribution by showing that *P. meridianus* occurs further southwest and northeast than previously described. It also illustrates that its range is much narrower from northwest to southeast than was illustrated prior (Figure 4).

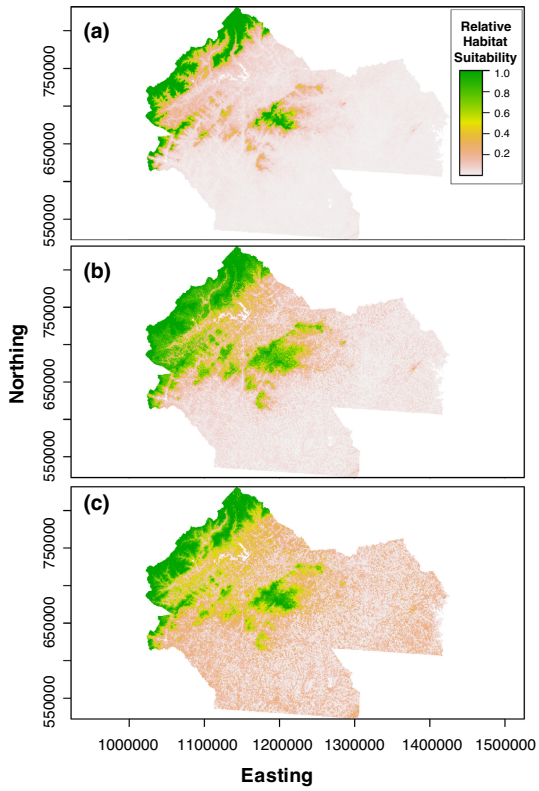

FIGURE 3 (a) Model b, built using database presences, is projected across the greater study area. (b) Model b_{all}, built using all available presence data. (c) Model b_{survey}, built using only survey data. Studies of salamander habitat using ecophysiology and landscape ecology have found that montane *Plethodon* species may persist near streams at their lower elevation range edge, as near-stream sites are often cooler and more humid (e.g., Gade & Peterman, 2019). The heat map produced by Model b_{all} is consistent with this inference, as high suitability areas appear as thin lines overlapping streams near the low elevation range edge. Coordinates use the North Carolina State Plane System

4 | DISCUSSION

In this study, we have shown how species distribution modelling methods can be used to better understand the distributions of species with few sampling records. Among methods, performance was mixed (Table 1; Question 1). Elevation exerted a large influence on SDMs, consistent with previous SDM research (Bourg et al., 2005; Oke & Thompson, 2015) and analyses of *Plethodon* distributions (Hairston, 1949; Kozak & Wiens, 2010). Model predictions exhibited less spatial autocorrelation in the interior of the range than outside the minimum convex polygon. This is likely due to the more extreme topographic relief in the centre of the range, whereas the range exterior is characterized by a gradual transition to lower elevations. Negative spatial autocorrelation inferred by three variants of the ESM suggests pathological inference. Spatial filtering had small impact on SDM performance within any given SDM technique (Table 1). We hypothesize the highly parametric nature of ESMs facilitates overfitting, explaining high omission errors and poor discrimination ability compared to Maxent. However, a difficulty in comparing ESMs to other methods is that ESMs can hypothetically be built in an infinite number of ways (e.g., different algorithms, different thresholds for retaining bivariate models, different model evaluation schemes), and it was not our goal to explore the entire universe of possibilities for ESMs. Instead, we simply used two different model criteria (AUC

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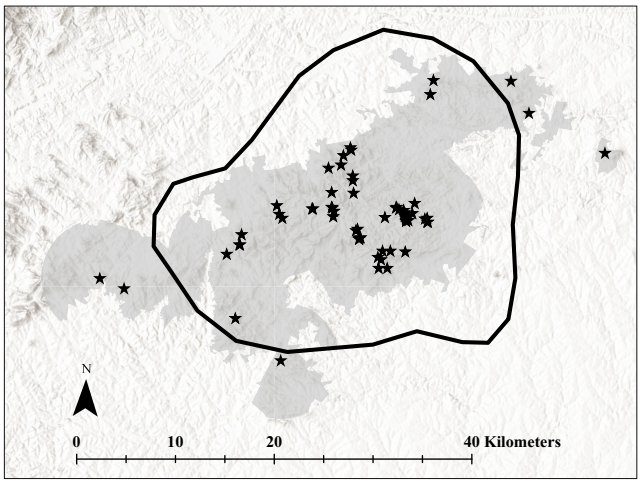

FIGURE 4 The IUCN range map of *Plethodon meridianus* (black outline) is compared with the new range map we propose (in grey). Stars are all presences in this study

and Boyce Index) and two of the most successful SDM algorithms for building ESMs (artificial neural networks and boosted regression trees; Breiner et al., 2018). Nearly all models performed well based on discrimination-based statistics (AUC); using a threshold-dependent statistic (OR), most methods performed well at predicting presences in the range centre, though all performed poorly at predicting populations in the range exterior (Table 1). There was little evidence for either agreement or a tradeoff in SDM predictions in the range interior versus exterior (Question 2). Our use of a distance raster as a simple, competing hypothesis was useful for showing that many methods did not have especially strong predictive ability (Table 1, Question 3). Presence data available from public databases were indeed from a biased region of environmental space (Figure 2, Question 4). We believe this may explain why an absolute (i.e., thresholded) measure of model performance (OR) indicated poor model performance in the range exterior while a relative measure of model performance (AUC) indicated good model performance (Table 1). Using different presence data sets to build SDMs using one of the best-performing methods (Model b_{all} versus b_{survey} or b) resulted in identifying different variable importance and regularization multipliers (Question 5). Increasing the number of presences (b_{all} vs. b_{survey}) did not lead to greater evenness among variable importance. This illustrates that sampling biases greatly shape how the underlying SDM is built, as the survey presences represent more geographically peripheral locations and, on average, lower elevations (Figure 2).

Because our conclusions about the performance of SDM methods are drawn from a single species, we cannot assume that our results generalize across species with few documented presences until additional case studies are conducted, yet we reiterate that this is the first comprehensive comparison of SDM methods given low sample sizes. When feasible, more systematic field surveys using stratified designs will offer stronger tests of SDM predictions (e.g., Rosner-Katz et al., 2020), though this practice is best reserved for species with high detection probabilities so that missed detections have a minimal impact on inferences. Additionally, researchers could simulate occupancy data for species with few records across an array of scenarios (e.g., dispersal limitation, competitive exclusion) and compare SDM techniques under each scenario.

A secondary goal of this study was to develop an improved range map for *P. meridianus*. A recent genetic analysis found that the westernmost population in our survey data could be in a hybrid zone with *P. amplus* or *P. metcalfi* (Heidler, 2020). The western terminus of the range may be a point where gene flow is actively merging species, or conversely where hybridization causes the range limit (Goldberg & Lande, 2007).

Why is the range of *P. meridianus* so restricted? Prior to conducting field surveys, it was predicted that the range of *P. meridianus* may be limited by competition with its widespread congener, *P. cylindraceus*. Yet *P. cylindraceus* was infrequently encountered in field surveys. We know of no mutualists that could potentially limit the range of *P. meridianus*. The suitable habitat lies outside the geographic range of *P. meridianus* (Figure 3a–c), indicative of dispersal limitation, but this habitat is occupied by closely related species that are also limited to high elevation habitats (*P. amplus*, *P. metcalfi* and *P. montanus*). While the South Mountains have experienced wildfire and logging (Smith, 1982), *P. meridianus* was seen in areas that have been logged and burned (Radomski in prep.), and thus, its detectability is unlikely to be influenced by human activity. If the geographic range is restricted due to biotic interactions, dispersal limitation, or demographic constraints, SDMs may underpredict the geographic range. Based on the survey data in this study, one might argue that *P. meridianus* appears to be geographically restricted as an artefact of limited sampling. However, even with the addition of survey presences, the geographic distribution of *P. meridianus* remains small (Figure 4). One could state that further sampling might result in finding *P. meridianus* in additional disjunct areas. Nevertheless, we believe further sampling would only result in minor changes to our current range map. Our confidence lies in the fact that biogeographic field research on salamanders of eastern North America is informed by high-density field sampling taking place over many decades (e.g., Hairston, 1949; Highton, 1995).

A limitation of SDMs is that they do not incorporate the potential correlation of environment and detection probability, so many areas could contain *P. meridianus* but individuals were simply not detected (Mackenzie et al., 2017). If detection probability negatively covaries with the same factors that affect habitat suitability, then SDMs may underpredict species' geographic ranges, or misidentify covariates that predict the geographic range. Riddell and Sears (2015) encountered fewer *P. metcalfi* when vapour pressure deficits were higher (drier conditions), typical of the low elevation sites they visited. In the southern Appalachian Mountains, lower elevations are often warmer and drier than higher elevations. Thus, conditions at low elevations may discourage salamanders from leaving their burrows, leading to missing observations. However, detection issues associated with elevation may be ameliorated in this study. First, subsequent visits in 2020 found the same lower elevational limit at the east-central range margin as in 2019, and each of the three

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independent sets of surveys found similar lower elevation limits. Second, the range of elevations that our data come from is roughly half that of Riddell and Sears (2015). Thus, the detection probability may not vary as much among sites we visited.

A priority for the future should be to assess how well different SDM techniques transfer in time to facilitate accurate forecasts of environmental changes, specifically when there is scarce presence data. While previous work has indicated that geographic distributions of species with small ranges are easier to model (McPherson & Jetz, 2007; Segurado & Araújo, 2004; Stockwell & Peterson, 2002), it is unclear whether those models transfer accurately. Model transferability is an imperfect practice, usually underestimating species' distributions since niches will almost always be under-characterized in the calibration region (Owens et al., 2013; Qiao et al., 2019; Saupe et al., 2012; Veloz et al., 2012). Transferability is hindered by temporal or spatial differences in species traits (e.g., local adaptation), different limiting factors across space or time and the changing correlation structure among environmental variables (Dormann et al., 2012). SDM transfer could be further complicated by biotic interactions. Nonetheless, model transferability remains key to forecasting the future distributions of species with few known presences.

In summary, various implementations of Maxent performed well at predicting the distribution of *P. meridianus*, a poorly known and geographically restricted salamander, but ESMs had higher omission error rates. Spatial filtering had very small impacts on Maxent models, and it actually decreased model performance for ESMs. Several techniques performed worse than using distance alone to predict presences. To ascertain generality, we suggest that researchers make similar comparisons of SDM techniques for species with few records. We also recommend that geographic distance be a standard against which to compare SDMs to evaluate their utility.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data and code used for analyses are available in the supporting information, which are available in Dryad (https://doi.org/10.5061/dryad.8931zcrsq).

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BIOSKETCH

The authors are interested in the natural history of amphibians, particularly salamanders. Most of their work research is conducted in Appalachia and the coastal plain, both of which are amphibian biodiversity hotspots.

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